NBA Head-to-Head Win Probability Prediction

Executive Summary — Erdős Institute Data Science Boot Camp, Summer 2026

Team: Jason Qin, Marshall Scott, Pengfei Zhang

Project Mentor: Alec Traaseth

Objectives

Develop a machine learning system that predicts NBA head-to-head win probability using historical team-level win/loss records and rolling individual/team performance metrics.

Stakeholders

General managers, coaches, prediction markets, sport betters, and gamblers — all users who benefit from a data-driven probability of a team winning a given matchup.

Data Collection & Cleaning

Import the play-by-play and box-score data via the nba_api package and open Kaggle datasets (historical box scores, player stats). Cleaned to remove non-competitive games (All-Star Weekend games) while retaining legitimate international regular-season games. Verified team-ID stability across franchise renames. Corrected missing game scores. Split data into Regular-Season-only and Regular+Playoff scopes for separate evaluation.

Exploratory Data Analysis

Computed scheduling context features (rest days, back-to-backs). Engineered home_team - minus away_team differential features from rolling averages (past 10 games) and exponentially-weighted (half-life = 3) team statistics: scoring, shooting efficiency, rebounding, assists, turnovers, steals, blocks, plus-minus, and recent win-rate. Visualizations confirmed these differentials meaningfully separated wins from losses. Across every model family tested, plusMinusPoints consistently emerged as the single strongest predictor, with recent win-rate and shooting efficiency close behind; scheduling features (rest days, back-to-backs) carried comparatively little predictive weight in the GAM and tree-based models.

Modeling

The base model, which was used as a reference, predicted that the home team would always win. This model had am accuracy of 57%. Two model families, encompassing five different models, were compared using season-by-season backtesting, and walk-forward cross-validation (rolling 5-year training window) across Regular Season and Combined datasets:

- Logistic Regression models
  - Logistic Regression (L1/LASSO) — interpretable baseline with directly calibrated probabilities.
  - Generalized Additive Models (GAMs) — Logistic regression + spline + tensor-interaction terms to capture non-linear effects.
- Ensemble models
  - Random Forest & Extra Trees — bagged ensembles for feature interactions and variance reduction.
  - XGBoost — gradient-boosted trees for sequential, loss-optimized predictions.

Final Results

Logistic Regression reached 64.8% accuracy (ROC-AUC 0.699) on the combined regular-season-plus-playoff dataset using the full engineered feature set. The Generalized Additive Model performed comparably at 62.4% accuracy (ROC-AUC 0.675) on its held-out test split, confirming that the relationships in the data are close to linear rather than sharply non-linear. Tree-based ensembles matched this range: Random Forest reached 64.9% accuracy in its best validation season, and Extra Trees peaked at 65.7% accuracy in 2024. XGBoost achieved 63.3% accuracy (ROC-AUC 0.674, log loss 0.645) on its chronological holdout. All models show an improvement over the base model.

Taken together, all five models converge to a similar ceiling of roughly 63–66% regular-season accuracy (ROC-AUC ≈ 0.65–0.71), pointing to an irreducible "any given night" noise floor in NBA outcomes rather than a modeling shortfall.

Future Directions

Several extensions could push predictive performance beyond the current ceiling:

- Player-level statistics — incorporating individual player efficiency, usage rate, and matchup-specific performance rather than relying solely on team-level aggregates.
- Injury reports — accounting for player availability and load management, which can shift a team's effective strength independent of its recent record.
- Lineup and rotation data — modeling how specific five-man lineups perform together, rather than treating a team as a single averaged unit.
- Betting market odds as a feature or benchmark — comparing model output against market-implied probabilities to gauge relative edge.
- Richer playoff-specific modeling — given the high variance observed in postseason-only backtests, a dedicated playoff model (or series-aware features, such as prior head-to-head results within the same series) may generalize better than transferring regular-season models directly.